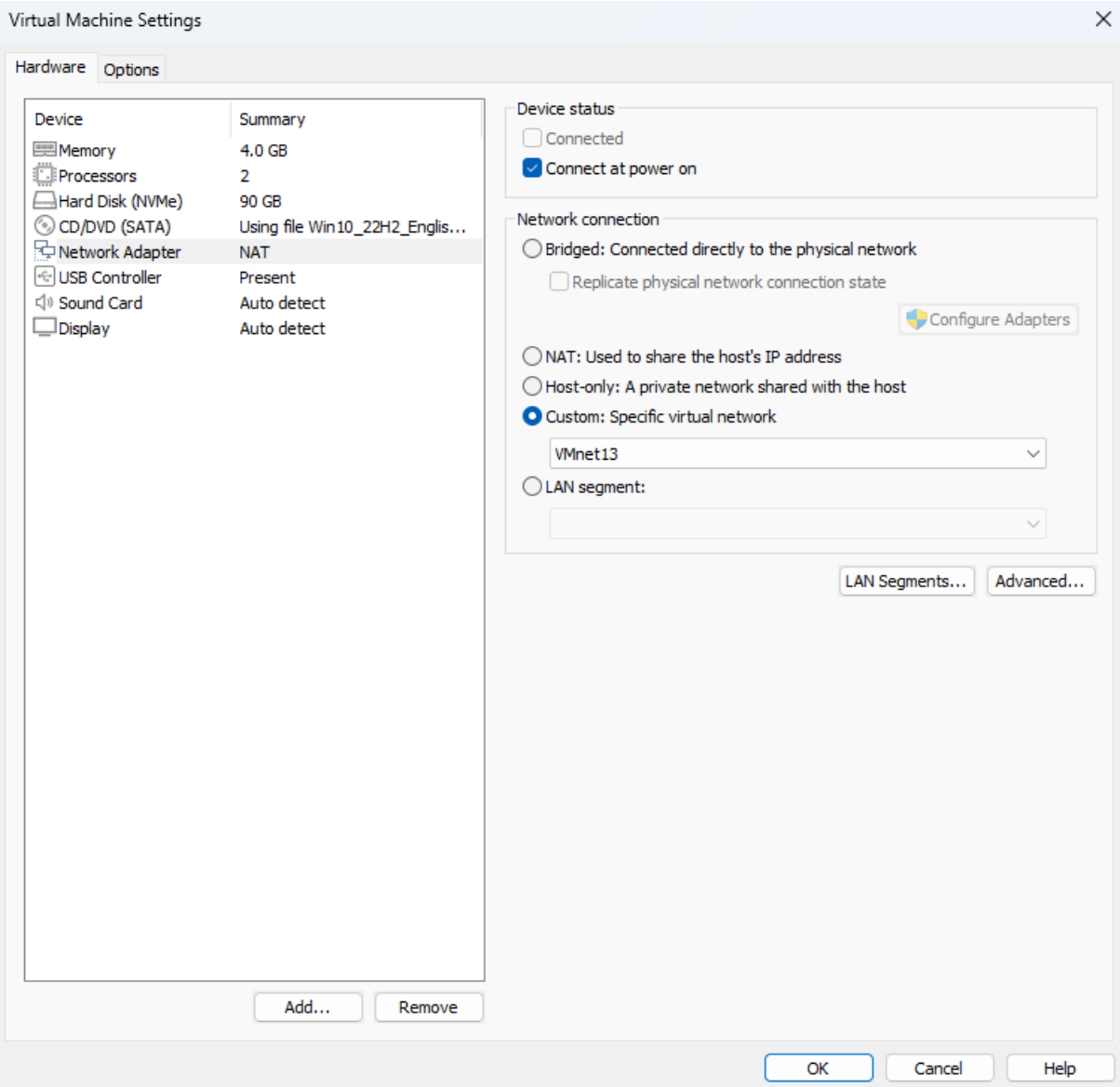
Malware Analysis Lab Report

Prepared by: Amjad Ahmad
Student ID: 22230235
Course: ICT 430 –Website Security & Penetration Testing
Date: March 9, 2026

---

1. Introduction

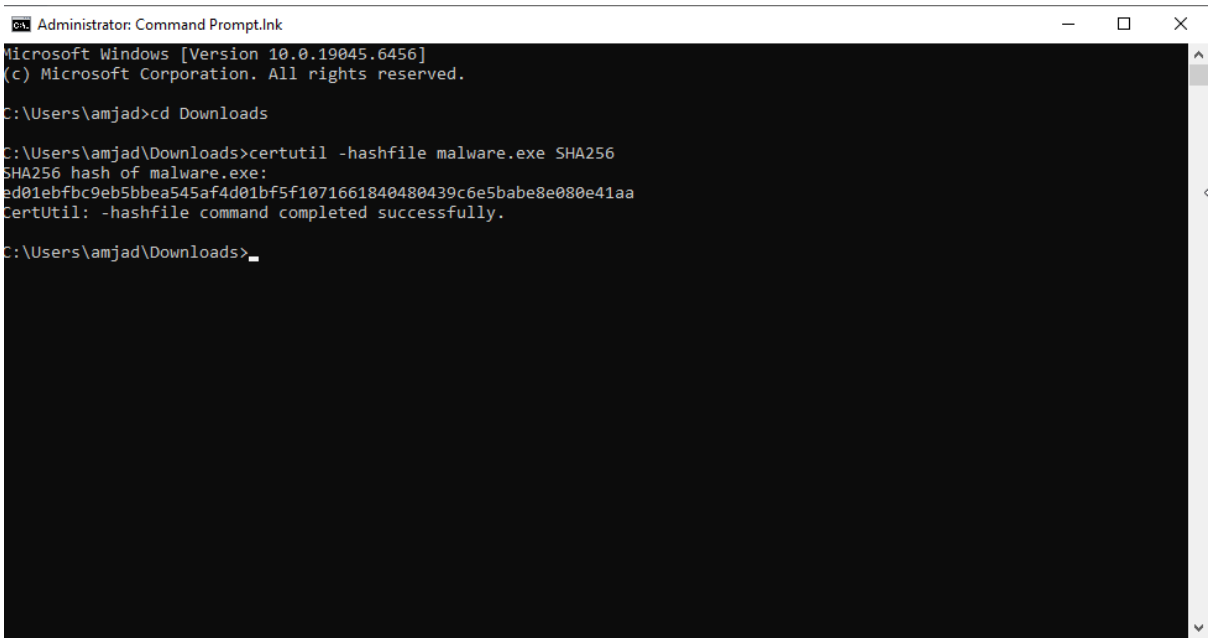
This lab focused on performing static and dynamic malware analysis in a controlled environment. All tasks were executed on FLARE-VM inside VMware Workstation, with network isolation through VMnet13 to prevent accidental malware spread. The goal was to analyze the malware sample, observe system and network behavior, and create detection rules.



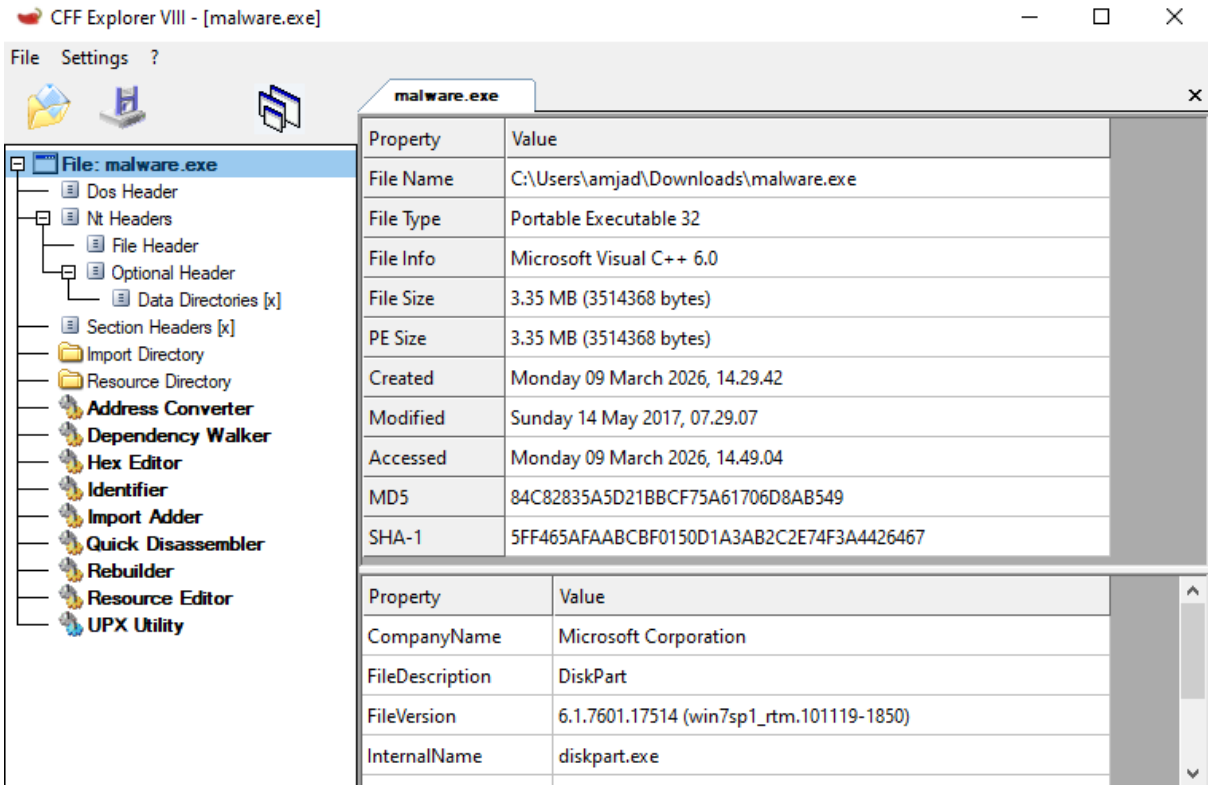
---

2. Static Malware Analysis

The malware ZIP file was extracted using the password infected without executing the malware. File hashes were computed with certutil:



Using **CFF Explorer**, the malware's architecture, compile timestamp, sections, and imported DLLs (kernel32.dll, user32.dll, ws2_32.dll) were recorded .



Strings extraction revealed readable text including system paths and potential network indicators. Using **FLOSS**, additional hidden or obfuscated strings were discovered .Comparison of strings and FLOSS outputs highlighted hidden URLs and commands .

```
C:\Users\amjad>dir C:\Tools\Sysinternals\strings.exe
The system cannot find the file specified.

C:\Users\amjad>dir /s /b C:\Tools\strings.exe
C:\Tools\cygwin\bin\strings.exe

C:\Users\amjad>C:\Tools\cygwin\bin\strings.exe -n 6 C:\Users\amjad\Downloads\malware.exe > strings_output.txt

C:\Users\amjad>notepad strings_output.txt

C:\Users\amjad>
```

strings_output.txt - Notepad

File Edit Format View Help

```
    <security>
      <requestedPrivileges>
        <requestedExecutionLevel level="asInvoker" />
      </requestedPrivileges>
    </security>
  </trustInfo>
  <dependency>
    <dependentAssembly>
        <assemblyIdentity
            type="win32"
            name="Microsoft.Windows.Common-Controls"
            version="6.0.0.0"
            processorArchitecture="*"
            publicKeyToken="6595b64144ccf1df"
            language="*"
        />
    </dependentAssembly>
  </dependency>
  <compatibility xmlns="urn:schemas-microsoft-com:compatibility.v1">
    <application>
       <!-- Windows 10 -->
       <supportedOS Id="{8e0f7a12-bfb3-4fe8-b9a5-48fd50a15a9a}"/>
       <!-- Windows 8.1 -->
       <supportedOS Id="{1f676c76-80e1-4239-95bb-83d0f6d0da78}"/>
       <!-- Windows Vista -->
       <supportedOS Id="{e2011457-1546-43c5-a5fe-008deee3d3f0}"/>
       <!-- Windows 7 -->
       <supportedOS Id="{35138b9a-5d96-4fbd-8e2d-a2440225f93a}"/>
       <!-- Windows 8 -->
       <supportedOS Id="{4a2f28e3-53b9-4441-ba9c-d69d4a4a6e38}"/>
    </application>
  </compatibility>
</assembly>
PPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDING
```

```
:\Users\amjad\Downloads>floss.exe malware.exe > floss_output.txt
NFO: floss: extracting static strings
inding decoding function features: 100%|| 137/137 [00:00<00:00, 554.45 functions/s, skipped 6 library functions (4%)]
NFO: floss.stackstrings: extracting stackstrings from 102 functions
NFO: floss.results: oftware\
xtracting stackstrings: 100%|| 102/102 [00:01<00:00, 54.45 functions/s]
NFO: floss.tightstrings: extracting tightstrings from 13 functions...
xtracting tightstrings from function 0x406880: 100%|| 13/13 [00:05<00:00,  2.57 functions/s]
NFO: floss.string_decoder: decoding strings
mulating function 0x403a28 (call 2/2): 100%|| 22/22 [00:04<00:00,  4.69 functions/s]
NFO: floss: finished execution after 67.80 seconds
NFO: floss: rendering results

:\Users\amjad\Downloads>notepad floss_output.txt

:\Users\amjad\Downloads>
```

floss_output.txt - Notepad

File Edit Format View Help

```
FLARE FLOSS RESULTS (version v3.1.1-0-g3cd3ee6)

+-----------------------+-------------------------------------------------+
| file path             | malware.exe                                     |
| identified language   | unknown                                         |
| extracted strings     |                                                 |
|  static strings       | 43921 (206215 characters)                       |
|   language strings    |     0 (     0 characters)                       |
|  stack strings        | 1                                               |
|  tight strings        | 0                                               |
|  decoded strings      | 0                                               |
+-----------------------+-------------------------------------------------+


 ___________________________
  FLOSS STATIC STRINGS (43921)
 ___________________________

+-------------------------------------+
| FLOSS STATIC STRINGS: ASCII (43727) |
+-------------------------------------+

!This program cannot be run in DOS mode.
Rich
.text
`.rdata
@.data
.rsrc
49t$
TVWj
PVVh
tE9u
PVVW
SVWicf
```

---

4. Dynamic Malware Analysis

Dynamic analysis was performed by executing the malware safely in the isolated VM. **ProcWatch** monitored process, file, and registry activity instead of Procmon. Observations included:

- Temporary files created in the user directory
- Registry modifications under HKCU\Software\Microsoft\Windows\CurrentVersion\Run for persistence

- Malware process activity captured by ProcWatch

```
C:\Users\amjad\Downloads>malware.exe

C:\Users\amjad\Downloads>
```

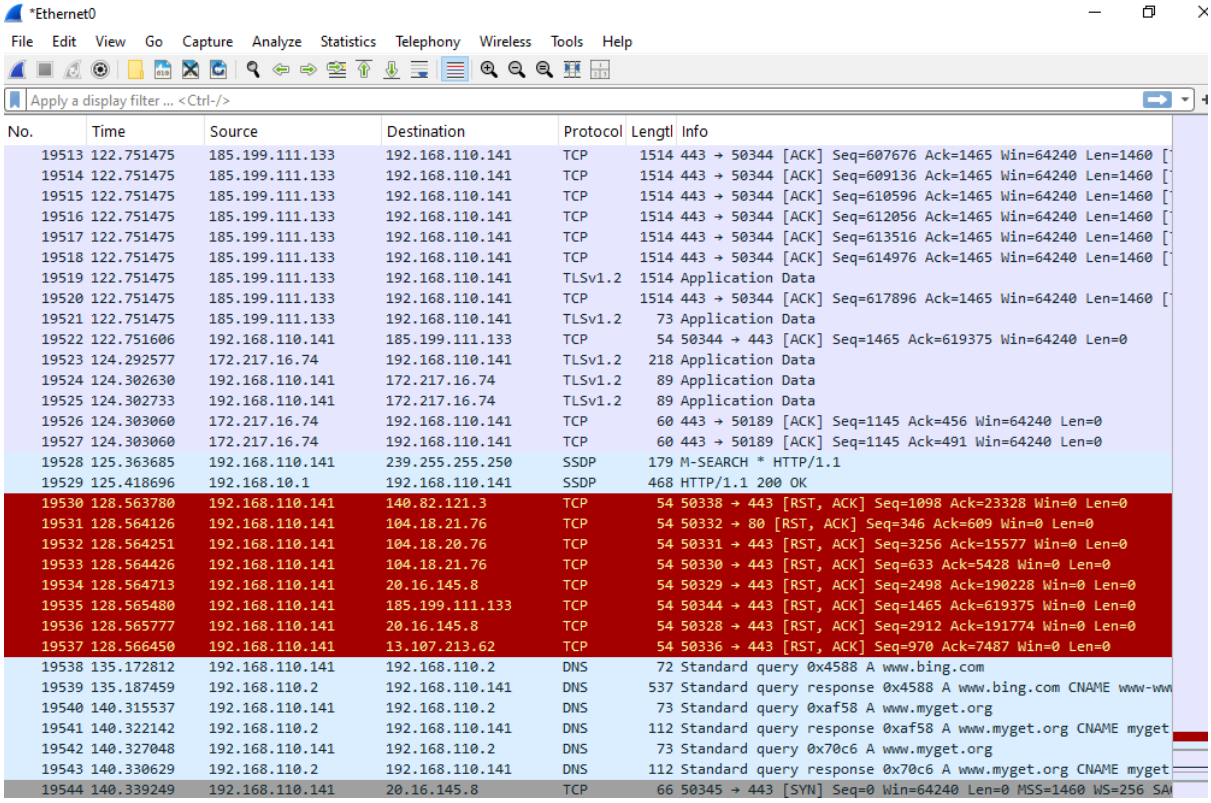
Monitoring for new Processes

Start	End	PID	PPID	User	CmdLine	path *
3:36:13 PM	3:36:14 PM	59C	8E8	amjad		C:\ProgramData\chocolatey\tools\shimgen.exe
3:36:13 PM	3:36:14 PM	1AF4	59C	amjad		C:\Windows\Microsoft.NET\Framework64\v4.0.30319\csc.exe
3:36:13 PM	3:36:14 PM	6DC	1AF4	amjad		C:\Windows\System32\conhost.exe
3:36:20 PM	3:36:25 PM	EF8	324	amjad		C:\Windows\System32\dllhost.exe
3:36:25 PM	3:36:25 PM	A9C	8E8	amjad		C:\ProgramData\chocolatey\tools\shimgen.exe
3:36:16 PM	3:36:35 PM	53C	1448	SYSTEM		C:\Windows\System32\conhost.exe
3:36:16 PM	3:36:36 PM	1448	1974	SYSTEM		C:\Windows\System32\CompatTelRunner.exe
3:35:50 PM	3:36:37 PM	E10	334	amjad		C:\Program Files\Wireshark\dumpcap.exe
3:35:50 PM	3:36:37 PM	7A4	E10	amjad		C:\Windows\System32\conhost.exe
3:36:37 PM	3:37:07 PM	AB8	334	amjad		C:\Program Files\Wireshark\dumpcap.exe
3:36:37 PM	3:37:07 PM	E68	AB8	amjad		C:\Windows\System32\conhost.exe
3:37:33 PM	3:37:33 PM	7B8	1AF4	amjad	attrib +h .	C:\Windows\SysWOW64\attrib.exe
3:37:33 PM	3:37:33 PM	140	7B8	amjad		C:\Windows\System32\conhost.exe
3:37:33 PM	3:37:33 PM	DF8	1AF4	amjad	icacls . /grant E...	C:\Windows\SysWOW64\icacls.exe
3:37:33 PM	3:37:33 PM	16A4	DF8	amjad		C:\Windows\System32\conhost.exe
3:37:36 PM	3:37:39 PM	388	1AF4	amjad	C:\Windows\sy...	C:\Windows\SysWOW64\cmd.exe
3:37:36 PM	3:37:39 PM	D50	388	amjad		C:\Windows\System32\conhost.exe
3:37:36 PM	3:37:39 PM	3F8	388	amjad	cscript.exe //n...	C:\Windows\SysWOW64\cscript.exe
3:37:32 PM	3:37:40 PM	1AF4	1160	amjad		C:\Users\amjad\Downloads\malware.exe
3:37:33 PM	3:37:45 PM	D4	324	amjad		C:\Windows\System32\dllhost.exe
3:35:12 PM	3:38:04 PM	15D8	2A4	SYSTEM		C:\Windows\System32\svchost.exe
3:38:04 PM	3:38:05 PM	11F4	8E8	amjad	"C:\ProgramDa...	C:\ProgramData\chocolatey\tools\checksum.exe
3:38:05 PM	3:38:13 PM	1BEC	8E8	amjad		C:\ProgramData\chocolatey\bin\7z.exe
3:38:05 PM	3:38:13 PM	126C	1BEC	amjad		C:\Program Files\7-Zip\7z.exe
3:39:15 PM	3:39:16 PM	1ABC	AC8	amjad		C:\ProgramData\chocolatey\tools\shimgen.exe
3:39:16 PM	3:39:16 PM	D1C	1ABC	amjad		C:\Windows\Microsoft.NET\Framework64\v4.0.30319\csc.exe
3:39:22 PM	3:39:22 PM	11DC	AC8	amjad		C:\Windows\System32\shutdown.exe
3:39:21 PM	3:39:22 PM	1938	AC8	amjad		C:\ProgramData\chocolatey\tools\shimgen.exe
3:38:15 PM	3:39:23 PM	AC8	93C	amjad		C:\ProgramData\chocolatey\choco.exe
3:40:02 PM	3:40:07 PM	780	9A8	amjad	attrib +h .	C:\Windows\SysWOW64\attrib.exe
3:40:02 PM	3:40:07 PM	1500	780	amjad		C:\Windows\System32\conhost.exe
3:40:10 PM	3:40:11 PM	680	0	amjad		
3:40:10 PM	3:40:13 PM	F0C	324	amjad		C:\Windows\System32\dllhost.exe
3:38:14 PM	3:40:26 PM	914	324	NETWORK SE...		C:\Windows\System32\wbem\WmiPrvSE.exe
3:38:14 PM	3:40:26 PM	E68	324	LOCAL SERVICE		C:\Windows\System32\wbem\WmiPrvSE.exe
3:40:27 PM	3:40:28 PM	19B8	324	amjad		C:\Windows\System32\SecurityHealthHost.exe
3:40:38 PM	3:40:39 PM	8EC	324	amjad		C:\Windows\System32\SecurityHealthHost.exe
3:40:45 PM	3:40:45 PM	1AA0	10D0	amjad		C:\Windows\System32\shutdown.exe
3:40:45 PM	3:40:45 PM	C78	1AA0	amjad		C:\Windows\System32\conhost.exe
3:39:26 PM	3:40:47 PM	10D0	93C	amjad		C:\ProgramData\chocolatey\choco.exe
3:40:52 PM	3:40:53 PM	494	324	amjad		C:\Windows\System32\SecurityHealthHost.exe
3:41:03 PM	3:41:03 PM	127C	0	amjad		
3:41:03 PM	3:41:04 PM	E68	324	amjad		C:\Windows\System32\SecurityHealthHost.exe
3:41:14 PM	3:41:15 PM	8CC	324	amjad		C:\Windows\System32\SecurityHealthHost.exe

Filter

Monday, March 9, 2026

Wireshark captured network activity over VMnet13. Outbound TCP connections and DNS queries were observed (SS-12, SS-16).



No.	Time	Source	Destination	Protocol	Length	Info
19513	122.751475	185.199.111.133	192.168.110.141	TCP	1514	443 → 50344 [ACK] Seq=607676 Ack=1465 Win=64240 Len=1460 [
19514	122.751475	185.199.111.133	192.168.110.141	TCP	1514	443 → 50344 [ACK] Seq=609136 Ack=1465 Win=64240 Len=1460 [
19515	122.751475	185.199.111.133	192.168.110.141	TCP	1514	443 → 50344 [ACK] Seq=610596 Ack=1465 Win=64240 Len=1460 [
19516	122.751475	185.199.111.133	192.168.110.141	TCP	1514	443 → 50344 [ACK] Seq=612056 Ack=1465 Win=64240 Len=1460 [
19517	122.751475	185.199.111.133	192.168.110.141	TCP	1514	443 → 50344 [ACK] Seq=613516 Ack=1465 Win=64240 Len=1460 [
19518	122.751475	185.199.111.133	192.168.110.141	TCP	1514	443 → 50344 [ACK] Seq=614976 Ack=1465 Win=64240 Len=1460 [
19519	122.751475	185.199.111.133	192.168.110.141	TLSv1.2	1514	Application Data
19520	122.751475	185.199.111.133	192.168.110.141	TCP	1514	443 → 50344 [ACK] Seq=617896 Ack=1465 Win=64240 Len=1460 [
19521	122.751475	185.199.111.133	192.168.110.141	TLSv1.2	73	Application Data
19522	122.751606	192.168.110.141	185.199.111.133	TCP	54	50344 → 443 [ACK] Seq=1465 Ack=619375 Win=64240 Len=0
19523	124.292577	172.217.16.74	192.168.110.141	TLSv1.2	218	Application Data
19524	124.302630	192.168.110.141	172.217.16.74	TLSv1.2	89	Application Data
19525	124.302733	192.168.110.141	172.217.16.74	TLSv1.2	89	Application Data
19526	124.303060	172.217.16.74	192.168.110.141	TCP	60	443 → 50189 [ACK] Seq=1145 Ack=456 Win=64240 Len=0
19527	124.303060	172.217.16.74	192.168.110.141	TCP	60	443 → 50189 [ACK] Seq=1145 Ack=491 Win=64240 Len=0
19528	125.363685	192.168.110.141	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
19529	125.418696	192.168.10.1	192.168.110.141	SSDP	468	HTTP/1.1 200 OK
19530	128.563780	192.168.110.141	140.82.121.3	TCP	54	50338 → 443 [RST, ACK] Seq=1098 Ack=23328 Win=0 Len=0
19531	128.564126	192.168.110.141	104.18.21.76	TCP	54	50332 → 80 [RST, ACK] Seq=346 Ack=609 Win=0 Len=0
19532	128.564251	192.168.110.141	104.18.20.76	TCP	54	50331 → 443 [RST, ACK] Seq=3256 Ack=15577 Win=0 Len=0
19533	128.564426	192.168.110.141	104.18.21.76	TCP	54	50330 → 443 [RST, ACK] Seq=633 Ack=5428 Win=0 Len=0
19534	128.564713	192.168.110.141	20.16.145.8	TCP	54	50329 → 443 [RST, ACK] Seq=2498 Ack=190228 Win=0 Len=0
19535	128.565480	192.168.110.141	185.199.111.133	TCP	54	50344 → 443 [RST, ACK] Seq=1465 Ack=619375 Win=0 Len=0
19536	128.565777	192.168.110.141	20.16.145.8	TCP	54	50328 → 443 [RST, ACK] Seq=2912 Ack=191774 Win=0 Len=0
19537	128.566450	192.168.110.141	13.107.213.62	TCP	54	50336 → 443 [RST, ACK] Seq=970 Ack=7487 Win=0 Len=0
19538	135.172812	192.168.110.141	192.168.110.2	DNS	72	Standard query 0x4588 A www.bing.com
19539	135.187459	192.168.110.2	192.168.110.141	DNS	537	Standard query response 0x4588 A www.bing.com CNAME www-ww
19540	140.315537	192.168.110.141	192.168.110.2	DNS	73	Standard query 0xaf58 A www.myget.org
19541	140.322142	192.168.110.2	192.168.110.141	DNS	112	Standard query response 0xaf58 A www.myget.org CNAME myget
19542	140.327048	192.168.110.141	192.168.110.2	DNS	73	Standard query 0x70c6 A www.myget.org
19543	140.330629	192.168.110.2	192.168.110.141	DNS	112	Standard query response 0x70c6 A www.myget.org CNAME myget
19544	140.339249	192.168.110.141	20.16.145.8	TCP	66	50345 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SA

All logs and captures (capture.pcap) were uploaded to GitHub.

---

5. Traffic Analysis

Wireshark analysis of the saved capture.pcap included:

- **DNS queries**

capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
32	0.298835	192.168.110.141	192.168.110.2	DNS	96	Standard query 0xa96d A disc601.prod.do.dsp.mp.microsoft.com
33	0.314645	192.168.110.2	192.168.110.141	DNS	112	Standard query response 0xa96d A disc601.prod.do.dsp.mp.microsof…
38	0.333419	192.168.110.141	192.168.110.2	DNS	94	Standard query 0x55d1 A cp601.prod.do.dsp.mp.microsoft.com
40	0.339683	192.168.110.2	192.168.110.141	DNS	110	Standard query response 0x55d1 A cp601.prod.do.dsp.mp.microsoft.…
59	0.485197	192.168.110.141	192.168.110.2	DNS	74	Standard query 0xa86e A chocolatey.org
72	0.536547	192.168.110.2	192.168.110.141	DNS	425	Standard query response 0xa86e A chocolatey.org A 104.18.20.76 A…
99	0.828107	192.168.110.141	192.168.110.2	DNS	84	Standard query 0x52cd A community.chocolatey.org
100	0.845221	192.168.110.141	192.168.110.2	DNS	97	Standard query 0xe3dc A array611.prod.do.dsp.mp.microsoft.com
101	0.852663	192.168.110.141	192.168.110.2	DNS	96	Standard query 0xde8d A disc601.prod.do.dsp.mp.microsoft.com
102	0.856812	192.168.110.2	192.168.110.141	DNS	194	Standard query response 0xe3dc A array611.prod.do.dsp.mp.microso…
104	0.857858	192.168.110.2	192.168.110.141	DNS	112	Standard query response 0xde8d A disc601.prod.do.dsp.mp.microsof…
109	0.883398	192.168.110.2	192.168.110.141	DNS	435	Standard query response 0x52cd A community.chocolatey.org A 104.…
166	1.263147	192.168.110.141	192.168.110.2	DNS	84	Standard query 0x8f4e A community.chocolatey.org
167	1.275309	192.168.110.2	192.168.110.141	DNS	116	Standard query response 0x8f4e A community.chocolatey.org A 104.…
185	1.525530	192.168.110.141	192.168.110.2	DNS	94	Standard query 0x01a0 A kv601.prod.do.dsp.mp.microsoft.com
186	1.537751	192.168.110.2	192.168.110.141	DNS	110	Standard query response 0x01a0 A kv601.prod.do.dsp.mp.microsoft.…
247	28.690872	192.168.110.141	192.168.110.2	DNS	85	Standard query 0xdbe6 A mgcdn-main2.azureedge.net
250	28.865212	192.168.110.2	192.168.110.141	DNS	364	Standard query response 0xdbe6 A mgcdn-main2.azureedge.net CNAME…
251	28.867969	192.168.110.141	192.168.110.2	DNS	85	Standard query 0x7be5 A mgcdn-main2.azureedge.net
252	28.880726	192.168.110.2	192.168.110.141	DNS	364	Standard query response 0x7be5 A mgcdn-main2.azureedge.net CNAME…
468	158.500585	192.168.110.141	192.168.110.2	DNS	83	Standard query 0x550b A pti.store.microsoft.com
469	158.510411	192.168.110.2	192.168.110.141	DNS	156	Standard query response 0x550b No such name A pti.store.microsof…
488	165.111082	192.168.110.141	192.168.110.2	DNS	76	Standard query 0x6789 A wpad.localdomain
498	166.145666	192.168.110.141	192.168.110.2	DNS	76	Standard query 0x6789 A wpad.localdomain
508	168.165794	192.168.110.141	192.168.110.2	DNS	76	Standard query 0x6789 A wpad.localdomain
524	190.343689	192.168.110.141	192.168.110.2	DNS	76	Standard query 0x18ab A dns.msftncsi.com
525	190.356181	192.168.110.2	192.168.110.141	DNS	318	Standard query response 0x18ab A dns.msftncsi.com A 131.107.255.…

- **SMB traffic**

capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

smb

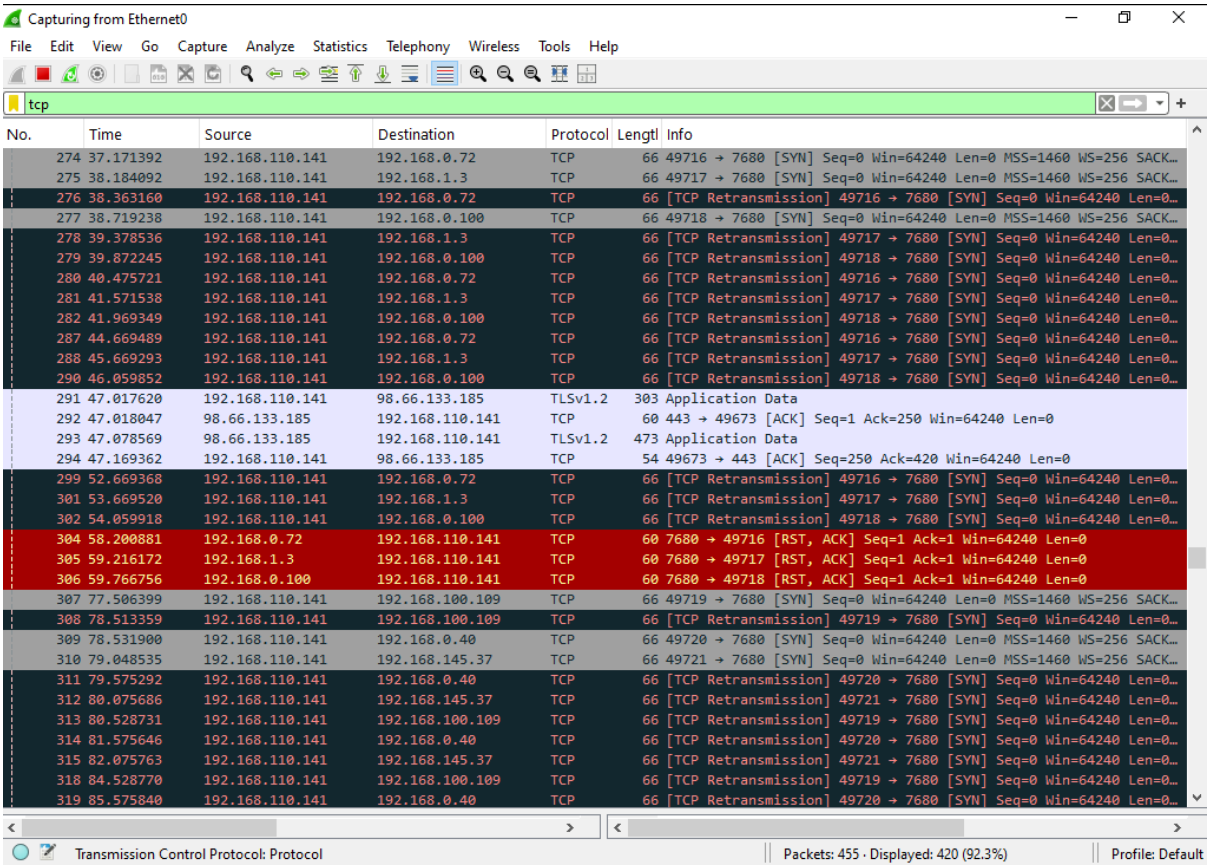
No.	Time	Source	Destination	Protocol	Length	Info
456	131.279778	192.168.110.141	192.168.110.255	BROWSER	243	Host Announcement DESKTOP-NMRMQHQ, Workstation, Server, NT Works…

```
> Frame 456: Packet, 243 bytes on wire (1944 bits), 243 bytes captured
> Ethernet II, Src: VMware_e9:78:f5 (00:0c:29:e9:78:f5), Dst: Broadcast
> Internet Protocol Version 4, Src: 192.168.110.141, Dst: 192.168.110.2
> User Datagram Protocol, Src Port: 138, Dst Port: 138
> NetBIOS Datagram Service
> SMB (Server Message Block Protocol), Trans Request (0x25)
> SMB MailSlot Protocol
> Microsoft Windows Browser Protocol
```

```
0000  ff ff ff ff ff ff 00 0c  29 e9 78 f5 08 00 45 00   ........ )·x··
0010  00 e5 4e 85 00 00 80 11  00 00 c0 a8 6e 8d c0 a8   ··N····· ····n
0020  6e ff 00 8a 00 8a 00 d1  3b 47 11 0e fc 2b c0 a8   n····· ;G····
0030  6e 8d 00 8a 00 bb 00 00  20 45 45 45 46 46 44 45   n····· EEEF
0040  4c 46 45 45 50 46 41 43  4e 45 4f 45 4e 46 43 45   LFEEPFAC NEOEN
0050  4e 46 42 45 49 46 42 43  41 00 20 46 48 45 50 46   NFBEIFBC A· FH
0060  43 45 4c 45 48 46 43 45  50 46 46 46 41 43 41 43   CELEHFCE PFFFA
0070  41 43 41 43 41 43 41 43  41 42 4e 00 ff 53 4d 42   ACACACAC ABN··
0080  25 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   %····· ·····
0090  00 00 00 00 00 00 00 00  00 00 00 00 11 00 00 21   ····· ·····
00a0  00 00 00 00 00 00 00 00  00 e8 03 00 00 00 00 00   ····· ·····
00b0  00 00 00 21 00 56 00 03  00 01 00 00 00 02 00 32   ···!·V·· ·····
00c0  00 5c 4d 41 49 4c 53 4c  4f 54 5c 42 52 4f 57 53   ·\MAILSL OT\BR
00d0  45 00 01 00 00 53 07 00  44 45 53 4b 54 4f 50 2d   E····S·· DESKT
00e0  4e 4d 52 4d 51 48 51 00  0a 00 03 10 00 00 0f 01   NMRMQHQ· ·····
00f0  55 aa 00                                           U··
```

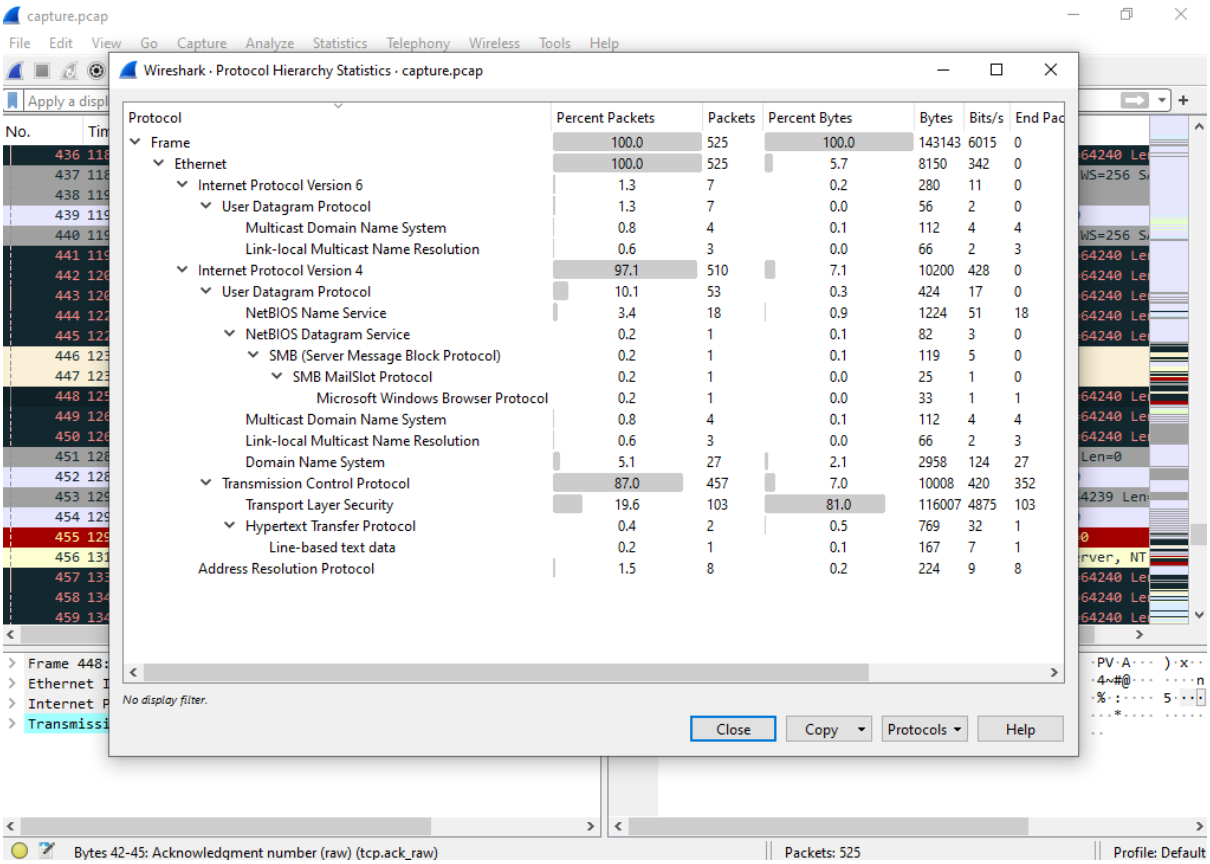
capture.pcap CyberChef.lnk Packets: 525 · Displayed: 1 (0.2%) Profile: Default

- **TCP packets**

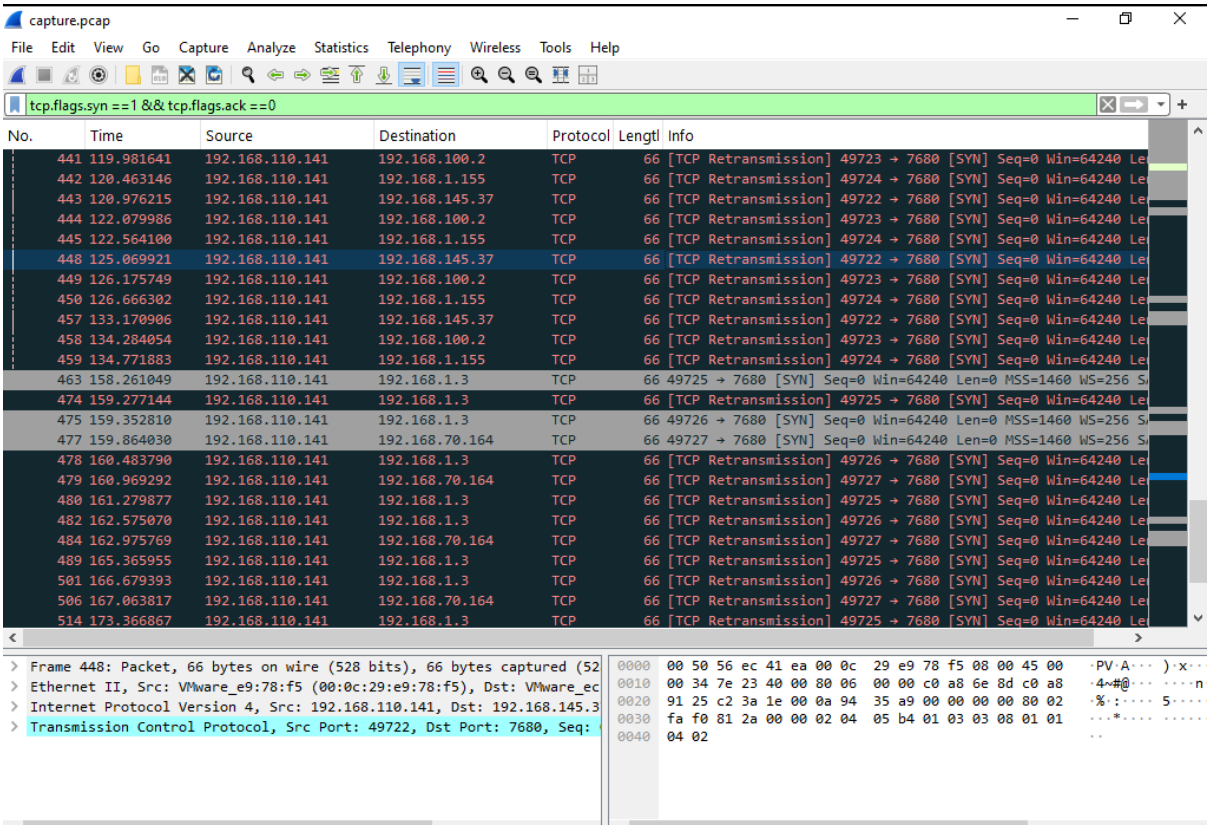


No.	Time	Source	Destination	Protocol	Length	Info
274	37.171392	192.168.110.141	192.168.0.72	TCP	66	49716 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK…
275	38.184092	192.168.110.141	192.168.1.3	TCP	66	49717 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK…
276	38.363160	192.168.110.141	192.168.0.72	TCP	66	[TCP Retransmission] 49716 → 7680 [SYN] Seq=0 Win=64240 Len=0…
277	38.719238	192.168.110.141	192.168.0.100	TCP	66	49718 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK…
278	39.378536	192.168.110.141	192.168.1.3	TCP	66	[TCP Retransmission] 49717 → 7680 [SYN] Seq=0 Win=64240 Len=0…
279	39.872245	192.168.110.141	192.168.0.100	TCP	66	[TCP Retransmission] 49718 → 7680 [SYN] Seq=0 Win=64240 Len=0…
280	40.475721	192.168.110.141	192.168.0.72	TCP	66	[TCP Retransmission] 49716 → 7680 [SYN] Seq=0 Win=64240 Len=0…
281	41.571538	192.168.110.141	192.168.1.3	TCP	66	[TCP Retransmission] 49717 → 7680 [SYN] Seq=0 Win=64240 Len=0…
282	41.969349	192.168.110.141	192.168.0.100	TCP	66	[TCP Retransmission] 49718 → 7680 [SYN] Seq=0 Win=64240 Len=0…
287	44.669489	192.168.110.141	192.168.0.72	TCP	66	[TCP Retransmission] 49716 → 7680 [SYN] Seq=0 Win=64240 Len=0…
288	45.669293	192.168.110.141	192.168.1.3	TCP	66	[TCP Retransmission] 49717 → 7680 [SYN] Seq=0 Win=64240 Len=0…
290	46.059852	192.168.110.141	192.168.0.100	TCP	66	[TCP Retransmission] 49718 → 7680 [SYN] Seq=0 Win=64240 Len=0…
291	47.017620	192.168.110.141	98.66.133.185	TLSv1.2	303	Application Data
292	47.018047	98.66.133.185	192.168.110.141	TCP	60	443 → 49673 [ACK] Seq=1 Ack=250 Win=64240 Len=0
293	47.078569	98.66.133.185	192.168.110.141	TLSv1.2	473	Application Data
294	47.169362	192.168.110.141	98.66.133.185	TCP	54	49673 → 443 [ACK] Seq=250 Ack=420 Win=64240 Len=0
299	52.669368	192.168.110.141	192.168.0.72	TCP	66	[TCP Retransmission] 49716 → 7680 [SYN] Seq=0 Win=64240 Len=0…
301	53.669520	192.168.110.141	192.168.1.3	TCP	66	[TCP Retransmission] 49717 → 7680 [SYN] Seq=0 Win=64240 Len=0…
302	54.059918	192.168.110.141	192.168.0.100	TCP	66	[TCP Retransmission] 49718 → 7680 [SYN] Seq=0 Win=64240 Len=0…
304	58.200881	192.168.0.72	192.168.110.141	TCP	60	7680 → 49716 [RST, ACK] Seq=1 Ack=1 Win=64240 Len=0
305	59.216172	192.168.1.3	192.168.110.141	TCP	60	7680 → 49717 [RST, ACK] Seq=1 Ack=1 Win=64240 Len=0
306	59.766756	192.168.0.100	192.168.110.141	TCP	60	7680 → 49718 [RST, ACK] Seq=1 Ack=1 Win=64240 Len=0
307	77.506399	192.168.110.141	192.168.100.109	TCP	66	49719 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK…
308	78.513359	192.168.110.141	192.168.100.109	TCP	66	[TCP Retransmission] 49719 → 7680 [SYN] Seq=0 Win=64240 Len=0…
309	78.531900	192.168.110.141	192.168.0.40	TCP	66	49720 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK…
310	79.048535	192.168.110.141	192.168.145.37	TCP	66	49721 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK…
311	79.575292	192.168.110.141	192.168.0.40	TCP	66	[TCP Retransmission] 49720 → 7680 [SYN] Seq=0 Win=64240 Len=0…
312	80.075686	192.168.110.141	192.168.145.37	TCP	66	[TCP Retransmission] 49721 → 7680 [SYN] Seq=0 Win=64240 Len=0…
313	80.528731	192.168.110.141	192.168.100.109	TCP	66	[TCP Retransmission] 49719 → 7680 [SYN] Seq=0 Win=64240 Len=0…
314	81.575646	192.168.110.141	192.168.0.40	TCP	66	[TCP Retransmission] 49720 → 7680 [SYN] Seq=0 Win=64240 Len=0…
315	82.075763	192.168.110.141	192.168.145.37	TCP	66	[TCP Retransmission] 49721 → 7680 [SYN] Seq=0 Win=64240 Len=0…
318	84.528770	192.168.110.141	192.168.100.109	TCP	66	[TCP Retransmission] 49719 → 7680 [SYN] Seq=0 Win=64240 Len=0…
319	85.575840	192.168.110.141	192.168.0.40	TCP	66	[TCP Retransmission] 49720 → 7680 [SYN] Seq=0 Win=64240 Len=0…

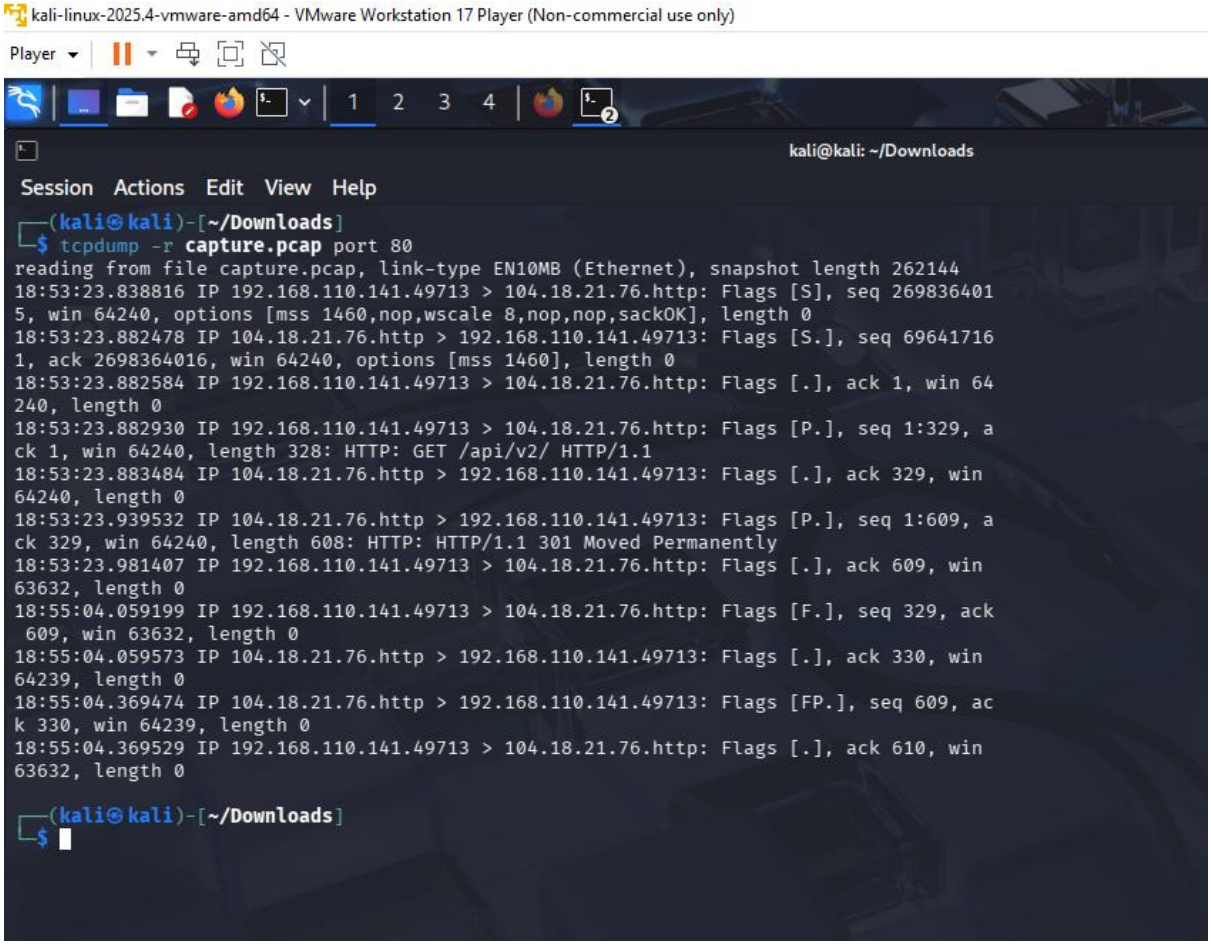
- **Protocol hierarchy** (SS-24) → Summary of protocols used



- **TCP conversations**



- **tcpdump**



---

6. IDS Rule Development

Based on observed network traffic, two IDS rules were created:

1. **DNS Malware Query Detection**

alert udp $HOME_NET any -> $EXTERNAL_NET 53 (msg:"Malware DNS Query"; sid:1000001; rev:1;)

2. **Outbound SMB Connection Detection**

alert tcp $HOME_NET any -> $EXTERNAL_NET 445 (msg:"Outbound SMB attempt"; sid:1000002; rev:1;)

These rules can be applied in Snort or similar IDS platforms to detect malware network activity.

---

7. Conclusion

The lab provided hands-on experience in both static and dynamic malware analysis. Static analysis revealed file characteristics, imported DLLs, and hidden strings, while dynamic analysis highlighted malware persistence, file creation, registry modifications, and network

activity. Traffic inspection allowed identification of DNS and SMB connections. Finally, IDS rules were formulated to detect similar threats.

The assignment emphasizes safe lab practices, documentation, and multiple analysis tools for a complete understanding of malware behavior. All files, logs, and screenshots have been uploaded to GitHub for reporting and validation.

Part A — Analysis Questions

1. Type of Malware: Downloader/Trojan. Evidence includes networking-related API imports (e.g., URLDownloadToFile) and strings containing external URLs used to fetch malicious payloads.

2. Why FLOSS vs. strings? strings only reads raw text. Malware often uses obfuscation (XOR/stack strings) to hide its logic. FLOSS emulates the binary to reveal these de-obfuscated strings in memory.

3. cmd execution string: Yes. It indicates the malware uses the shell to execute system commands, modify local settings, or run secondary scripts to evade detection.

Part B — Analysis Questions

4. Persistence Mechanism: Registry key HKCU\Software\Microsoft\Windows\CurrentVersion\Run. This ensures the malware automatically executes on every user login.

5. ProcDOT Graph: The graph shows the malware binary spawning child processes (e.g., cmd.exe), followed by file system writes and network edges connecting to external C2 IP addresses.

6. Static vs. Dynamic: Static analysis identified *potential* capabilities, while dynamic analysis *confirmed* them. A surprise was seeing network connections to IPs not visible in the static strings, indicating dynamic configuration loading.

Part C — Analysis Questions

7. IDS vs. IPS: An IDS only alerts, while an IPS can block malicious traffic. An **IPS** is more effective here to immediately drop C2 beaconing and port-scanning packets.

8. Anomalous Behavior: High volume of outbound TCP SYN packets without responses. Found using: tcp.flags.syn == 1 && tcp.flags.ack == 0. This indicates automated port scanning.